

Engineering Journal

Date 04/08/25

What was supposed to be accomplished today according to the Gantt Chart:

Today we officially began the project and collaborated to brainstorm initial product ideas related to heat exhaustion. As this was our first working session, our Gantt Chart was created during this time, so there were no prior assigned tasks yet.

Tatum

Work According to Gantt Chart:

Not applicable—Gantt Chart was created today.

Accomplishments and Investigations (with images if available):

- Participated in a team brainstorming session.
 - Researched the symptoms and risks of heat exhaustion in various environments (e.g., lifeguarding, hiking).
 - Helped frame key questions:
 - What population are we targeting?
 - What body region should be monitored (excluding the torso)?
 - What type of sensors would be most effective?
-

Conner

Work According to Gantt Chart:

Not applicable—Gantt Chart was created today.

Accomplishments and Investigations (with images if available):

- Looked up statistics related to heat exhaustion, especially in outdoor occupations and recreational activities.

- Suggested hiking backpacks with heat monitoring and cooling features.
 - Helped refine the problem statement and product criteria.
 - Researched existing wearable technologies and their limitations.
-

Alvaro

Work According to Gantt Chart:

Not applicable—Gantt Chart was created today.

Accomplishments and Investigations (with images if available):

- Contributed to the creation of the Gantt Chart to plan our next steps.
 - Brought up considerations around comfort, pre existing products, functionality
 - Participated in evaluating feasibility of visor, hat, and backpack ideas.
-

Are you on track with your Gantt Chart?

Yes—today was a partial planning day. We created the Gantt Chart and started strong with collaborative brainstorming and research.

Are there any key obstacles in the way moving forward?

Yes. The biggest current obstacle is narrowing down:

- **Target population** (lifeguards, hikers, outdoor workers, etc.)
- **Sensor selection** (must be non-invasive, reliable, and body-safe in high-heat environments)

How we plan to overcome them:

- Assign each team member to research one area: population needs, potential body regions, and available sensors.
- Begin initial concept sketches once key decisions are finalized.

Date: 04/10/25

What was supposed to be accomplished today according to the Gantt Chart:

Today's objective was to begin preliminary research on heat exhaustion, investigate the feasibility of our initial design ideas, and develop a draft medical need statement. These tasks were aimed at helping us narrow down a direction for our wearable concept.

Team Member: Tatum

Work According to Gantt Chart:

Conducted background research on heat exhaustion relevant to target populations.

Accomplishments and Investigations (with images if available):

- Researched heat exhaustion risks and statistics, with a focus on lifeguards and individuals exposed to prolonged sun.
 - Identified common symptoms and environmental conditions contributing to heat-related illnesses.
-

Team Member: Alvaro

Work According to Gantt Chart:

Explore design ideas

Accomplishments and Investigations (with images if available):

- Investigated the practicality of the lifeguard visor concept.
- Identified limitations related to power integration, size constraints, and water hazards around the head.
- Based on findings, contributed to team discussions about pivoting toward a more feasible product idea. Hikers?

Team Member: Conner

Work According to Gantt Chart:

Drafted the medical need statement. Explore design ideas

Accomplishments and Investigations (with images if available):

- Wrote a preliminary medical need statement
- Incorporated feedback from the team and began preparing for revisions based on our shift in direction to hikers.

Are you on track with your Gantt Chart?

Yes—we are on track. All members completed their assigned early-stage research and planning tasks. The team also made an important strategic pivot to improve the project's feasibility.

Are there any key obstacles in the way moving forward?

Yes.

- Initial limitations in the visor design led us to reassess the product scope.
- Choosing a final target population that provides both a strong use case and feasible design options is ongoing.

How we plan to overcome them:

- The team agreed to shift focus toward **hikers** as the target population due to the increased flexibility in wearables (e.g., hats, backpacks, straps).
- Next steps include researching sensors that can be integrated into hiking gear and evaluating common heat-related risks for hikers.
- We'll update the Gantt Chart accordingly to reflect this new direction and delegate tasks for deeper technical and market research.

Date 04/11/25

What was supposed to be accomplished today according to the Gantt Chart:

Today's focus was to continue refining the medical need statement, begin looking into sensor integration, and start early prototyping ideas, including evaluating potential wearable platforms like hats.

Team Member: Tatum

Work According to Gantt Chart:

Sourced literature to support the medical need and began testing wearable platforms.

Accomplishments and Investigations (with images if available):

- Found a relevant peer-reviewed source that supports the risk of heat exhaustion in outdoor activity populations: DOI: 10.3390/ijerph17114086
This study provides evidence on the physiological stress heat places on active individuals, strengthening our justification for a wearable heat monitoring device.
- Purchased a \$9.19 lightweight hat from Amazon to serve as a prototype platform for sensor integration.



Team Member: Alvaro

Work According to Gantt Chart:

Focused on technical development of core sensors.

Accomplishments and Investigations (with images if available):

- Worked on debugging the **MAX30102 pulse oximeter sensor**, which is key for tracking heart rate and possibly oxygen saturation as physiological indicators of heat stress.
 - Investigated signal noise and began validating sensor output using Arduino.
-

Team Member: Conner**Work According to Gantt Chart:**

Continued development of design documentation and refinement of problem definition.

Accomplishments and Investigations (with images if available):

- Updated the medical need statement to reflect a more specific target population and application setting (hikers and outdoor enthusiasts).
 - Began initial **design sketches** for the wearable, exploring layout and possible sensor placements on the hat.
-

Are you on track with your Gantt Chart?

Yes, we are currently on track. Key early-stage tasks are progressing as scheduled, including background research, sensor testing, and prototyping. We are beginning to transition into technical integration and user-focused design.

Are there any key obstacles in the way moving forward?

Not currently, but we plan to **check in with Dr. Gutruf** soon to validate our direction and receive feedback on feasibility, sensor integration, and overall concept refinement.

Date 04/15/25

What was supposed to be accomplished today according to the Gantt Chart:

Today's goals were to continue technical research on heat exhaustion indicators, identify and source necessary components, update sensor selections, and explore temperature measurement reliability in various environmental conditions.

Team Member: Tatum

Work According to Gantt Chart:

Continued research on physiological signs of heat exhaustion and initiated purchasing logistics.

Accomplishments and Investigations (with images if available):

- Confirmed that heat exhaustion occurs when **core body temperature reaches 104°F (40°C)** or higher.
Source: [Mayo Clinic – Heat Exhaustion](#)
- Researched heart rate and SpO₂ trends during heat exhaustion using peer-reviewed literature:
DOI: 10.1152/ajpregu.00190.2014
- Compiled and **emailed a purchase list** including key components:
 - **6-pin connector (right angle)**, Qty: 2
[S6B-PH-K-S from Digi-Key](#)
 - **2-pin connector (right angle)**, Qty: 1
[S2B-PH-K-S from Digi-Key](#)
 - **MAX30205** human body temperature sensor, Qty: 1
[MAX30205MTA+T from Digi-Key](#)
 - **LoRa module**, Qty: 1
(Low-power, long-range communication)

Team Member: Alvaro

Work According to Gantt Chart:

Worked on integration planning in **Eagle CAD** and reviewed sensor datasheets.

Accomplishments and Investigations (with images if available):

- Focused on circuit design involving the following sensors:
 - **MAX30102** (pulse oximeter):
[Datasheet – Digi-Key](#)
 - **MAX30205** (body temp):
[Datasheet – Analog Devices](#)
 - **LIS2DH12 Accelerometer** (for motion detection):
[Datasheet – STMicroelectronics](#)
 - Reviewed LoRa module specifications:
 - [Comimark LoRa SX1278](#) (Selected for long-range data communication from the wearable device)
-

Team Member: Conner

Work According to Gantt Chart:

Researched **forehead temperature accuracy** and its correlation to core body temperature.

Accomplishments and Investigations (with images if available):

Identified key articles and studies on temperature reliability in varying environments:

- Looked for research papers on accurately measuring forehead temperature in different environments
 - Relating core temperature to forehead temperature
 - [A Comparative Study of Forehead Temperature and Core Body Temperature under Varying Ambient Temperature Conditions - PMC](#)
 - Measuring forehead temperature in hot environments accurately
 - [An Advanced Internet of Things System for Heatstroke Prevention with a Noninvasive Dual-Heat-Flux Thermometer - PMC](#)
 - [Validity of Devices That Assess Body Temperature During Outdoor Exercise in the Heat - PMC](#)
 - [Forehead skin temperature and thermal sensation during exercise in cool and thermoneutral environments - PubMed](#)

- Core temperature and gait measurements used in coordination to predict heat stroke
 - [Gait instability and estimated core temperature predict exertional heat stroke](#)
 - Findings support that forehead skin temperature is **environment-dependent**, but useful when corrected with algorithms or used alongside motion data.
-

Are you on track with your Gantt Chart?

Yes. We're progressing steadily. We also made some **sensor selection adjustments** based on design considerations. We **removed the external temperature sensor** from our design to reduce complexity and are instead exploring the use of an **accelerometer** to help detect signs of fatigue or instability associated with heat exhaustion.

Date 04/17/25

What was supposed to be accomplished today according to the Gantt Chart:

Key goals for today included implementing LoRa communication, finalizing PCB design, beginning brochure design, and continuing sensor research and development.

Team Member: Tatum

Work According to Gantt Chart:

- Began developing **LoRa transmitter code** for wireless data communication
 - Started initial layout and design of the **project brochure**
-

Team Member: Alvaro

Work According to Gantt Chart:

- **Completed 1st attempt mainboard design**
 - Worked on **LoRa receiver code**, setting up communication with the transmitter
-

Team Member: Conner

Work According to Gantt Chart:

- Focused on **temperature sensor integration** into the system LTspice
 - Continued reading **academic papers** to optimize temperature sensing accuracy
-

Are you on track with your Gantt Chart?

Yes. We are on track, though the addition of **LoRa communication** required us to shift some time toward **testing and debugging**, which will continue into tomorrow.

Date 04/18/25

What was supposed to be accomplished today according to the Gantt Chart:

Today's tasks focused on preparing the system for hardware implementation and finalizing early-stage documentation and component selection.

Team Member: Tatum

Work According to Gantt Chart: Brochure design and lora testing

- Continued **designing the brochure**, adding visuals and component functions



- Worked on **LoRa communication code** refinement

Link to code

<https://docs.google.com/document/d/1pp0nQ-1nx7exKagIXM7CzrC26ITA6704dSGEqEy3Mc8/edit?usp=sharing>

- Began researching **gold standard devices** for heat exhaustion detection (forehead thermometers, pulse oximeters, and wearable monitors)

Sensor lab check out gold standards

- Astroskin headband
- Astroskin device accelerometer
- Withings Thermo Smart Tempora

Team Member: Alvaro

Work According to Gantt Chart: Print PCB and start steps to lead to assembly

- **Submitted mainboard** for printing
- Finalized **electronic components list** for integration

Components main board

- One 1.8V LDO (LN620618)
 - One 3.3V LDO (ME6211C33M5G-N)
 - One ESP-12F
 - Accelerometer (LIS2DH12TR)
 - Two MOSFET (BSS138)
 - Three 22 uF capacitors
 - Two 1uF capacitors
 - Six 220nF capacitors
 - Two 4.7uF capacitors
 - Seven 4.7 kOhms resistors
 - Five 10kOhms resistors
-

Team Member: Conner

Work According to Gantt Chart: finish temperature sensor and accumulating paper references

- Worked on **safety and effectiveness research** related to commercial temperature monitoring systems
 - Created temperature sensor on LTspice
 - Continued reading and compiling data to compare **sensor accuracy and placement**
-

Are you on track with your Gantt Chart?

Yes, we are on track. Each team member is progressing according to our updated timeline, especially with the addition of LoRa communication and brochure development.

Are there any key obstacles in the way moving forward?

One potential obstacle is ensuring **consistent and accurate data transmission with LoRa modules** in different environments.

- **Plan to overcome:** Continue testing LoRa indoors and outdoors, measure signal strength

Date: 04/22/25

What was supposed to be accomplished today according to the Gantt Chart:

Today's tasks focused on continued development of the LoRa communication system, PCB soldering, part selection for analog circuits, and early Wheatstone bridge integration for sensor calibration.

Team Member: Tatum

Work According to Gantt Chart:

- Finalized an **additional parts list** needed for circuit design:
 - **Quantity: 8** – 6 kΩ resistors
 - **Quantity: 1** – Operational Amplifier
 - **Quantity: 5** – 10 kΩ resistors
 - **Quantity: 1** – 4.02 kΩ resistor
 - **Quantity: 1** – 220 nF capacitor
 - **Quantity: 1** – 22 μF capacitor
 - **Quantity: 1** – 1 μF capacitor
- Continued **testing LoRa communication** between transmitter and receiver units to ensure reliable data transmission.

Team Member: Conner

Work According to Gantt Chart:

- Began **Wheatstone bridge creation** to integrate with temperature sensor design for enhanced accuracy
- Printed a **custom PCB board** for integration
- Referenced a helpful guide for LoRa communication setup:
[LoRa Sender & Receiver Communication Guide](#)

Team Member: Alvaro

Work According to Gantt Chart:

- Soldering on new board with more broad channels, beginning with resistors and capacitors
- Assisted in **troubleshooting and refining LoRa communication system** during collaborative testing

Are you on track with your Gantt Chart?

Yes. We're progressing steadily with both hardware integration and communication setup. LoRa testing is ongoing. We have ran into some issues with this aspect of the project but are continuing to work though simultaneously completing other tasks to stay on track.

Date: 04/24/25

What was supposed to be accomplished today according to the Gantt Chart:

Today's focus included further development of LoRa communication hardware, circuit simulations, pulse oximeter housing, and enclosure prototyping.

Team Member: Alvaro

Work According to Gantt Chart:

- Continued **soldering the ESP module** for integration with other components
 - Worked on the **LoRa receiver setup** for reliable data capture and transmission testing
-

Team Member: Conner Smith

Work According to Gantt Chart:

- Conducted **LTspice experiments** to model and validate the analog circuit for temperature measurement
 - Identified needed resistor values for next steps:
 - **Eight 4.02 kΩ resistors**
 - **Four 8.2 kΩ resistors**
 - **Four 18 kΩ resistors**
 - Began designing the **pulse oximeter enclosure**
-

Team Member: Tatum Abbruscato

Work According to Gantt Chart:

- Successfully **transmitted LoRa data for the first time**, receiving an “**AT, ok**” confirmation response

First successful code shown below

Transmitting code

```
void setup() {
  Serial.begin(115200); // Start Serial Monitor for debugging
  delay(2000);          // Allow LoRa module to power up

  // Set LoRa parameters for US region
  Serial.println("AT+BAND=915000000"); // Set frequency to 915
MHz
  delay(100);

  Serial.println("AT+ADDRESS=101"); // Set this device's
address
  delay(100);

  Serial.println("AT+NETWORKID=5"); // Set shared network ID
  delay(100);
}

void loop() {
  // Send message every 5 seconds
  Serial.println("AT+SEND=102,10,Hello ESP!"); // Send message
to address 102, with 10 bytes
  delay(5000); // Wait 5 seconds
}
```

Receiver code

```
void setup() {
  Serial.begin(115200); // Start Serial Monitor for debugging
(TX=GPI01, RX=GPI03)
  delay(2000);          // Allow LoRa module to power up

  // Set up receiver with correct frequency, network ID, and
address
  Serial.println("AT+BAND=915000000\r\n"); // 915 MHz for US
  delay(100);

  Serial.println("AT+ADDRESS=102\r\n"); // Set receiver's
address
}
```

```
delay(100);

Serial.println("AT+NETWORKID=5\r\n"); // Shared network ID
delay(100);
}

void loop() {
    // Check if data is available on the serial input
    if (Serial.available()) {
        String received = Serial.readString(); // Read incoming
data
        Serial.print("Received: ");
        Serial.println(received); // Print the received message to
the serial monitor
        delay(5000);
    }
}
```

- Developed the initial **LoRa box prototype**, including casing layout *had to reprint due to port accessibility issue



Are you on track with your Gantt Chart?

Majority yes — we've had to **adjust the Gantt Chart** to allow more time for testing and enclosure refinement.

Are there any key obstacles in the way moving forward?

Yes — the **LoRa box prototype's port wall height** is too short, which affects cable fit and usability.

How we're overcoming it:

We plan to **modify the 3D design and reprint** the enclosure with a properly measured port wall to ensure stable hardware connection.

Date: 04/25/25

What was supposed to be accomplished today according to the Gantt Chart:

Today's goals were to finalize PCB soldering, continue improving LoRa communication integration, and begin testing feedback elements like the LED and buzzer. Code development and hardware debugging were also planned.

Team Member: Conner Smith

Work According to Gantt Chart:

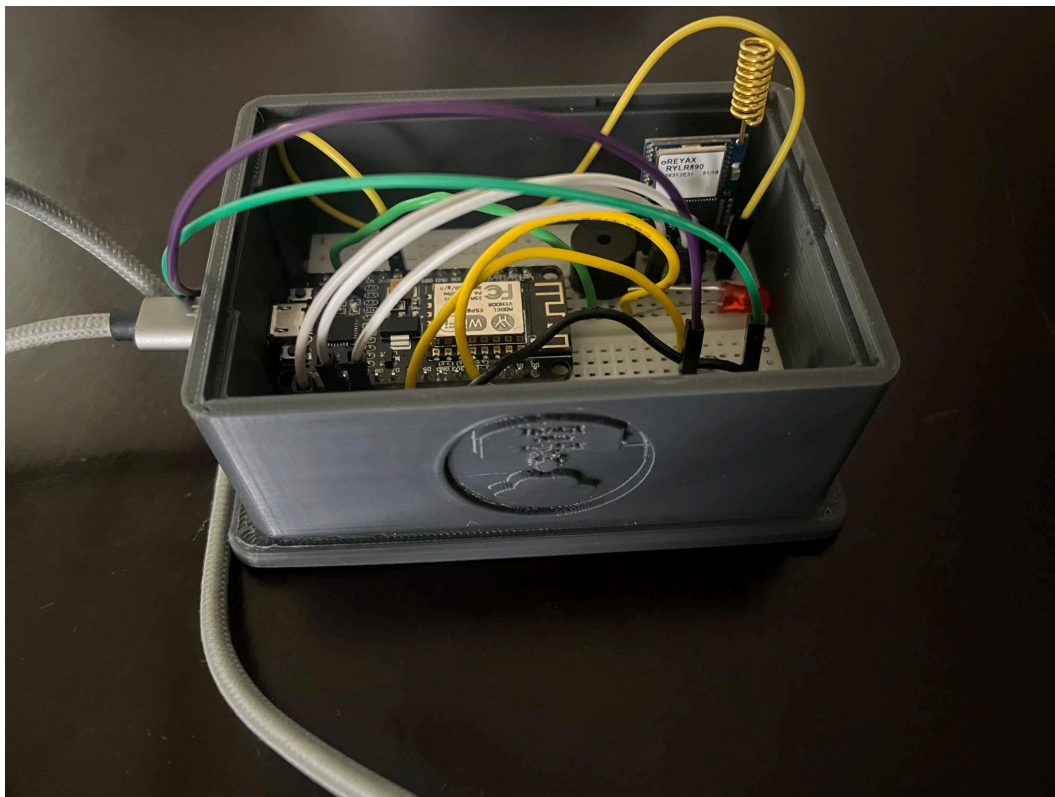
- Redesigned the **LoRa communication range station module** to include a **lid compatible with LED and buzzer** placement
 - Began **printing the pulse oximeter board** for circuit testing and eventual housing
-

Team Member: Tatum Abbruscato

Work According to Gantt Chart:

- Developed and tested **buzzer and LED alert code** for feedback signaling

Buzzer code shown below



Buzzer Blink Code

```
int buzzerPin = 5;    // D1 on ESP8266
int ledPin = 12;      // D6 on ESP8266

void setup() {
  pinMode(buzzerPin, OUTPUT);
  pinMode(ledPin, OUTPUT);
}

void loop() {
  digitalWrite(buzzerPin, HIGH); // Turn buzzer on
  digitalWrite(ledPin, HIGH);    // Turn LED on
  delay(75);                     // LED on for 150 ms

  digitalWrite(ledPin, LOW);     // Turn LED off
  delay(75);                     // LED off for 150 ms

  digitalWrite(buzzerPin, LOW);  // Turn buzzer off
  delay(300);                    // Wait before next beep
}
```

- Began **accelerometer code integration**
- Sent **accelerometer data via LoRa communication** for real-time transmission trials

Team Member: Alvaro

Work According to Gantt Chart:

- **Completed soldering** of the main PCB
- **Acquired new parts** to replace components from the previous board (which had faulty/bad channels)

- Began **probing circuits to check for correct voltage outputs**
-

Are you on track with your Gantt Chart?

Yes — everything is progressing smoothly and on schedule.

Are there any key obstacles in the way moving forward?

No major obstacles at this time.

How we're overcoming potential setbacks:

Any discrepancies in voltage or signal performance are being addressed immediately through probing and debugging. Replacing parts from faulty boards early helps maintain momentum.

Date: 04/29/25

What was supposed to be accomplished today according to the Gantt Chart:

Today's plan included further hardware development, soldering the pulse oximeter board, testing the accelerometer with a separate ESP, and making mechanical adjustments such as printing enclosures and logo patches.

Team Member: Conner Smith

Work According to Gantt Chart:

- **Printed new LoRa box** with updated edits for improved functionality and component fit
 - **Started soldering pulse oximeter board**
-

Team Member: Tatum Abbruscato

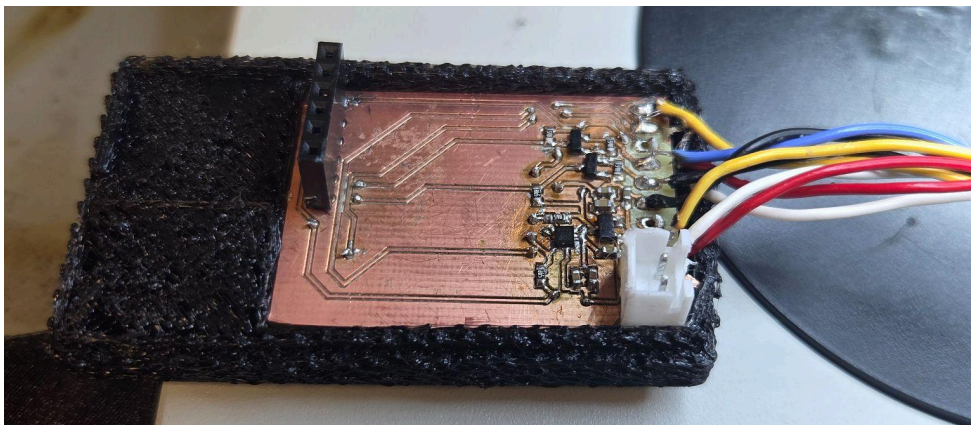
Work According to Gantt Chart:

- Worked with Alvaro to **debug the accelerometer using a separate ESP**

- Continued **development of accelerometer and pulse oximeter code**
- **Embroidered logo patch** for branding



- **Printed pulse oximeter enclosure** using TPU material for flexibility and durability



Team Member: Alvaro

Work According to Gantt Chart:

- Collaborated with Tatum to **debug the accelerometer setup with separate ESP**
- **Acquired replacement parts** for the old PCB (which had bad channels)
- **Began probing the new board for correct voltage outputs**

Are you on track with your Gantt Chart?

Mostly — we are progressing well, but **pulse oximeter board soldering** is slightly behind schedule.

Are there any key obstacles in the way moving forward?

- The team focused on **accelerometer debugging today**, which shifted time away from pulse oximeter tasks.

How we're overcoming this:

We **plan to attend office hours** to catch up on pulse oximeter soldering and stay on track for future milestones.

Date: 05/01/25

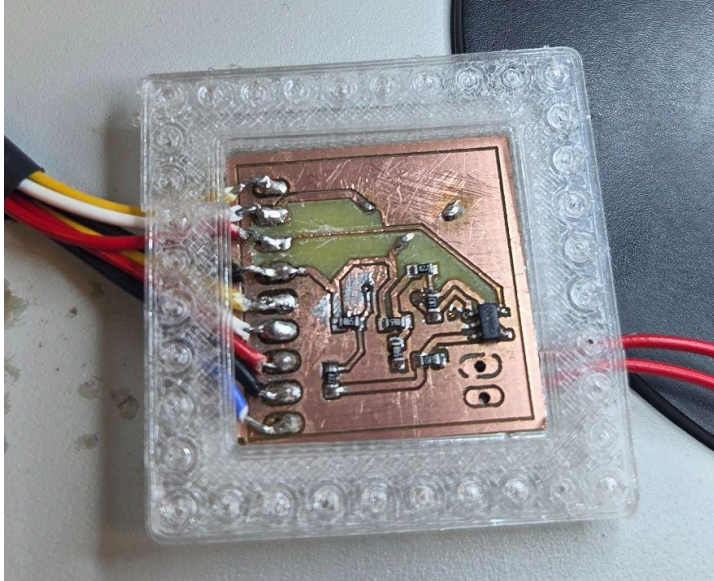
What Was Supposed to Be Accomplished Today (According to the Gantt Chart):

- **Finish sewing pocket on hat**
 - **Finish buzzer code**
 - **Create battery enclosure in preparation for hat integration**
-

Team Member: Conner Smith

Work According to Gantt Chart:

- Printed new LoRa box with updated edits
- Continued soldering pulse oximeter board
- Thermistor enclosure



Team Member: Tatum Abbruscato

Work According to Gantt Chart:

- Debugged accelerometer with separate ESP
- Continued development of accelerometer, temperature, and pulse oximeter code

- Sewed back pocket on hat



- Printed pulse oximeter enclosure with TPU for flexibility and durability

Team Member: Alvaro

Work According to Gantt Chart:

- Collaborated with Tatum to debug the accelerometer with separate ESP
 - Acquired new parts for the old PCB (with bad channels)
 - Began probing the new board for correct voltage outputs
-

Are You on Track with Your Gantt Chart?

- **Mostly on track**, but we need to catch up on **pulse oximeter soldering/debugging** as we focused on the accelerometer today.
 - **Progress is good**, and the **LoRa system is successfully transmitting all required data** using commercial boards.
-

Are There Any Key Obstacles in the Way Moving Forward?

- **Focus on accelerometer tasks** shifted attention away from the pulse oximeter.
 - **Decision pending** on whether to **commit to commercial sensors** for code development or continue debugging with custom sensors as the competition approaches.
-

How We're Overcoming These Obstacles:

- Plan to attend **office hours** to catch up on pulse oximeter soldering.
- Continue **resoldering the accelerometer** and **probing for shorts**, with input from **Gutruf** to ensure progress.
- **Decision on sensors** will be made soon to align with the upcoming competition deadline.

Date: 05/02/25

What was supposed to be accomplished today according to the Gantt Chart:

- Assemble hat and make finishing touches on the GUI display
-

Team Member: Conner Smith

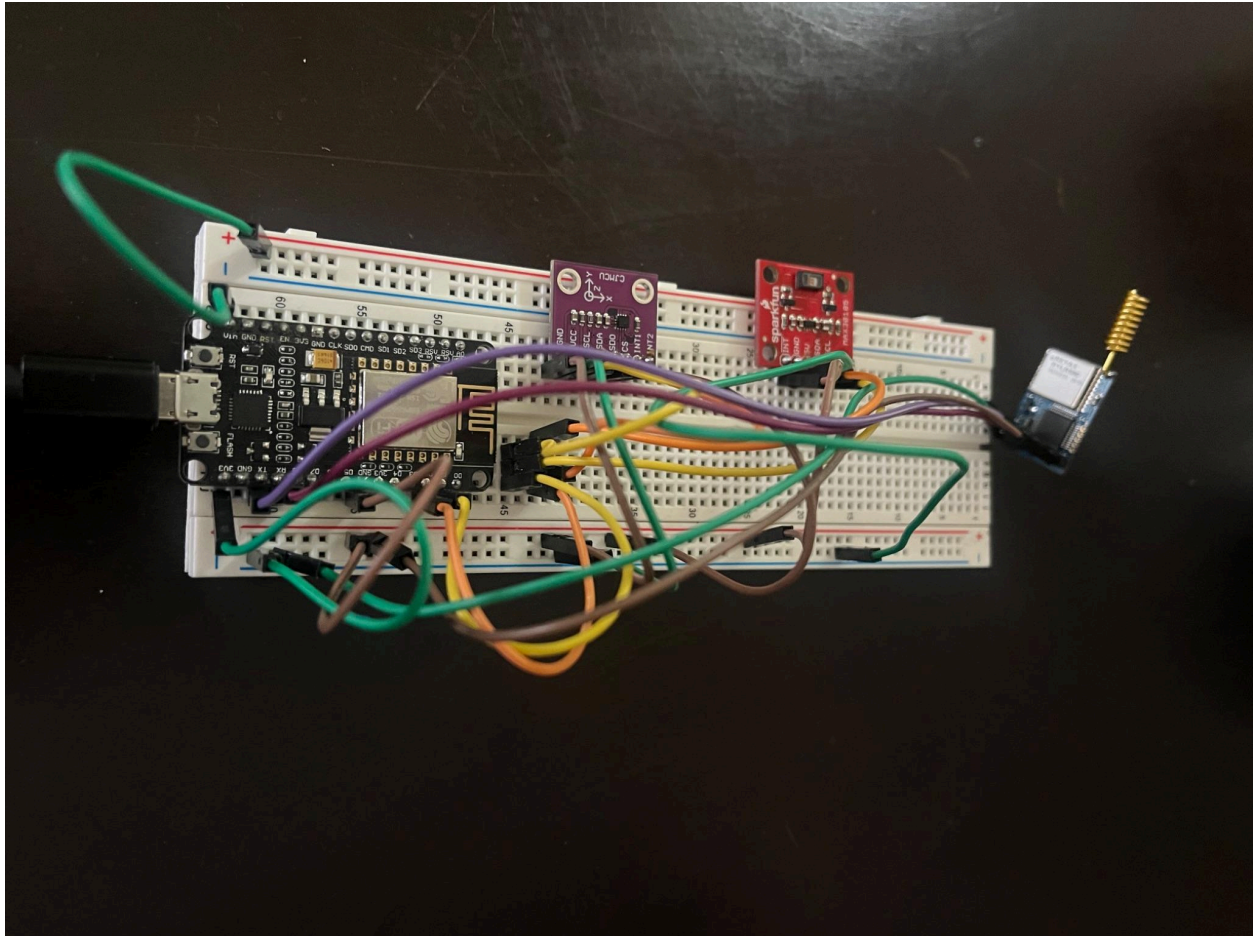
Work According to Gantt Chart:

- Work on report
 - Created an exploded view of the hat for assembly reference
 - Continued coding for the temperature sensor to ensure accurate readings
-

Team Member: Tatum Abbruscato

- Successfully outputted data for temperature, pulse oximeter, and accelerometer

Code for transmitting code shown below full data transmission



```
#include <Wire.h>
#include "MAX30105.h"
#include "heartRate.h"

#define LIS2DH_ADDR 0x18 // I2C address for LIS2DH

MAX30105 particleSensor;

const int ROLLING_WINDOW = 4;
float bpmBuffer[ROLLING_WINDOW];
int bpmIndex = 0;
int bpmSamples = 0;

float beatsPerMinute = 0;
float avgBPM = 0;
long lastBeat = 0;
float temperatureC = 0;
float temperatureF = 0;
```

```

unsigned long lastSendTime = 0;
const unsigned long sendInterval = 5000;  // Send every 5
seconds

void setup() {
  Serial.begin(115200);
  delay(2000);  // Allow LoRa module to power up

  Wire.begin(4, 5);  // SDA = GPIO4 (D2), SCL = GPIO5 (D1) on
ESP8266

  // === MAX30105 Init ===
  if (!particleSensor.begin(Wire, I2C_SPEED_STANDARD)) {
    Serial.println("MAX30105 not found. Check wiring.");
    while (1);
  }

  particleSensor.setup();
  particleSensor.setPulseAmplitudeRed(0x2F);
  particleSensor.setPulseAmplitudeIR(0x2F);
  particleSensor.setPulseAmplitudeGreen(0x00);

  Serial.println("MAX30105 initialized.");

  // === LIS2DH Init ===
  Wire.beginTransmission(LIS2DH_ADDR);
  Wire.write(0x20);  // CTRL_REG1
  Wire.write(0x57);  // Normal mode, all axes enabled
  Wire.endTransmission();

  Wire.beginTransmission(LIS2DH_ADDR);
  Wire.write(0x23);  // CTRL_REG4
  Wire.write(0x00);  // ±2g range
  Wire.endTransmission();

  delay(100);

  // === LoRa AT Config ===
  Serial.println("AT+BAND=915000000");
  delay(100);

```

```

Serial.println("AT+ADDRESS=101");
delay(100);
Serial.println("AT+NETWORKID=5");
delay(100);

Serial.println("Sensors initialized. Place your finger on the
MAX30105.");
}

void loop() {
    long irValue = particleSensor.getIR();
    bool fingerDetected = irValue > 5000;

    if (fingerDetected && checkForBeat(irValue)) {
        long currentTime = millis();
        long delta = currentTime - lastBeat;
        lastBeat = currentTime;

        float bpm = 60.0 / (delta / 1000.0);
        if (bpm > 40 && bpm < 180) {
            beatsPerMinute = bpm;
            bpmBuffer[bpmIndex++] = bpm;
            if (bpmIndex >= ROLLING_WINDOW) bpmIndex = 0;
            if (bpmSamples < ROLLING_WINDOW) bpmSamples++;

            avgBPM = 0;
            for (int i = 0; i < bpmSamples; i++) {
                avgBPM += bpmBuffer[i];
            }
            avgBPM /= bpmSamples;
        }
    }

    if (millis() - lastSendTime >= sendInterval) {
        int16_t x_raw = 0, y_raw = 0, z_raw = 0;

        Wire.beginTransmission(LIS2DH_ADDR);
        Wire.write(0x28 | 0x80);
        Wire.endTransmission();
        byte count = Wire.requestFrom(LIS2DH_ADDR, 6);
    }
}

```

```

if (count == 6) {
  x_raw = (Wire.read() | (Wire.read() << 8));
  y_raw = (Wire.read() | (Wire.read() << 8));
  z_raw = (Wire.read() | (Wire.read() << 8));

  float x_g = x_raw * 0.001;
  float y_g = y_raw * 0.001;
  float z_g = z_raw * 0.001;

  // === Read temperature and convert to F ===
  temperatureC = particleSensor.readTemperature();
  temperatureF = temperatureC * 9.0 / 5.0 + 32.0;

  // === Construct LoRa message ===
  String message = "x=" + String(x_g, 4) +
    ",y=" + String(y_g, 4) +
    ",z=" + String(z_g, 4) +
    ",TempF=" + String(temperatureF, 1);

  if (fingerDetected) {
    message += ",BPM=" + String(beatsPerMinute, 1) +
      ",AvgBPM=" + String(avgBPM, 1);
  } else {
    message += ",No contact with sensor detected";
  }

  int msgLength = message.length();
  Serial.println("AT+SEND=102," + String(msgLength) + "," +
message);
  } else {
    Serial.println("Failed to read accelerometer data.");
  }

  lastSendTime = millis();
}

delay(10);
}

```

Receiver

```

#include <Wire.h>

#define LIS2DH_ADDR 0x18 // I2C address for LIS2DH

void setup() {
  Serial.begin(115200); // Start Serial Monitor for debugging
  delay(2000);          // Allow LoRa module to power up

  // Set LoRa parameters for US region
  Serial.println("AT+BAND=915000000"); // Set frequency to 915
MHz
  delay(100);

  Serial.println("AT+ADDRESS=102"); // Set this device's
address (must be the same as sender's address)
  delay(100);

  Serial.println("AT+NETWORKID=5"); // Set shared network ID
(must match sender's network ID)
  delay(100);

  // LoRa module needs to be in receive mode
  Serial.println("AT+RXMODE=1"); // Put LoRa module in
receive mode
  delay(100);
}

void loop() {
  // Check if there is any incoming message
  Serial.println("AT+RECV"); // Check for new data in
receive buffer
  delay(1000);
}

```

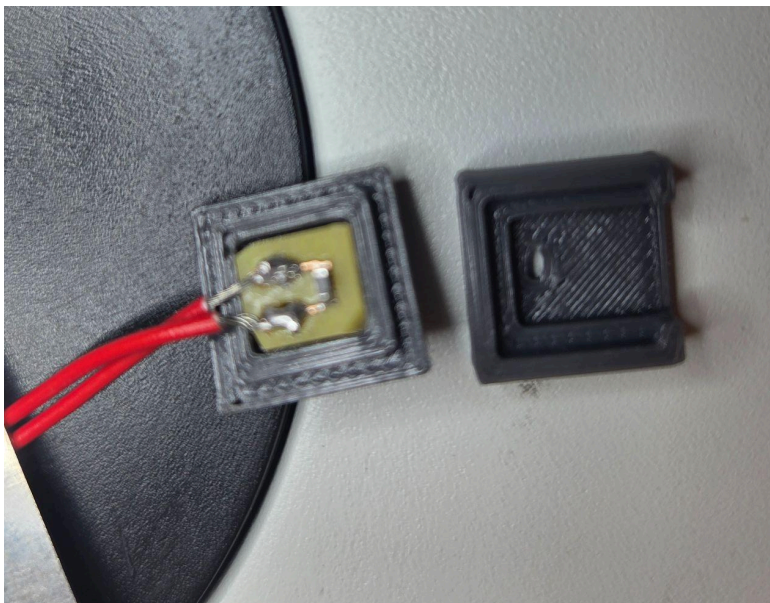
```
// If data is received, it will be printed here
if (Serial.available()) {
    String receivedMessage = Serial.readString(); // Read the
received data
    Serial.print("Received: ");
    Serial.println(receivedMessage); // Print the received
message
}

delay(1000); // Delay before checking for new data again
}
```

- Set monitoring thresholds to activate alerts when values exceed safe levels
 - Made improvements to the GUI display for clarity and ease of use
-

Team Member: Alvaro

- Assembled boards into a functional system
- Began debugging temperature sensor
- Worked on soldering pin attachments for better connectivity
- Started assembling enclosures for final system assembly





Are you on track with your Gantt Chart?

Partially — we are slightly behind on hat assembly and the soldering of the ESP with finished code, but we are making good progress on other aspects.

Temperature sensor is looking promising.

Obstacles Moving Forward:

- Hat assembly delays
 - Need to solder ESP and ensure code integration
 - Final system enclosure assembly
-

How We Plan to Overcome Them:

- Plan to attend office hours to troubleshoot code and hardware integration
- Focus on completing soldering and ESP setup during the weekend
- Use EDC time over the weekend to catch up on assembly tasks

Days attended office hours to catch up with gantt chart

- April 23
- April 28
- April 29
- May 4th

Met at the EDC

- May 2nd
- May 3rd
- May 5th

